MACHINE LEARNING TUTORIAL ASSIGNMENT

VISION TRANSFORMERS FOR SKIN LESION CLASSIFICATION USING TRANSFER LEARNING

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1. INTRODUCTION

Vision Transformers (ViTs) are deep learning models that apply Transformers, which have become foundational in natural language processing, to Computer vision tasks, by using the same principles used in text-based models. ViTs were first introduced in 2020 by Google Research in the paper **"An Image is Worth 16×16 Words,"[1]** showing that transformers trained on large image datasets can match or beat CNNs. Transformers entered vision to provide global context modeling through self-attention mechanism that can capture long-range spatial relationships.

In this tutorial, we explore the application of Vision Transformers to the problem of skin lesion classification (HAM10000 dataset) using transfer learning approach.

2. VISION TRANSFORMERS

- Vision Transformers processes an image as a sequence of data, like large language model processes words in a sentence.
- It breaks images into patches (eg:16x16 or 32x32 pixels), like cutting a photo into tiles, then using attention mechanism to understand how different parts relate to each other in a global context.
- ViTs use a pure transformer encoder architecture along with patch embedding, CLS token and MLP Classification Head.

Architecture – Components

1. **Image Patching**
   - The image is divided into smaller, fixed size patches. In the original proposed paper, the patch size is 16x16. This means that for a 224x224 image, if patch size is 16x16, then image splits into 14x14 = 196 patches (224/16=14).
   - Therefore, the shape of a patch would be **16x16x3,** (3- RGB color channels), making each patch into a 3-dimentional tensor.
   - Each patch contains localized information (spatial and color from a local region) like word tokens in transformers.

2. **Patch Embedding**
   - Each image patch (16x16 RGB) is converted into numerical representation, which is known as embedding.
   - Every patch is flattened into a single vector, that means unrolling all pixels and color values (16x16x3 = 768) values into single line numbers.
   - These are passed through a **linear projection layer**, which aligns all patch tokens into learnable suitable for a transformer encoder.
3. **Positional Encoding**
   - Transformers treat sequences of inputs as unordered, as the processes in parallel, all at once approach.
   - But it is crucial to know the position of a patch that represents a specific region of the original image.
   - To resolve this, positional encoding is added to each patch embedding (a vector - that acts like a tag for positional information).
   - ViTs use **Learned Positional encoding,** which are trainable parameters – a model learns during training. There are various other positional encoding techniques used in other advanced ViT variants.
4. **Transformer Encoder Block**
   - The embedded patches are passed through the Transformer Encoder Block. It consists of the following components:
   - **Multi-Head Attention**

Self-attention mechanism helps the model to determine what parts of the image are most relevant to each other. Initially for each input patch, three vectors are created: **Query(Q), Key(K), Value(V).**

Query -Key interaction involves processing current input as **Query** which relates to all other inputs**, Keys** to obtain attention scores. These are scaled and passed through a softmax, turning them into attention weights, then taking a weighted sum of **Value** vectors.

$$\textbf{Attention(Q,K,V)} = softmax\left(\frac{\left(QK^T\right)}{\sqrt{dk}}\right) V$$

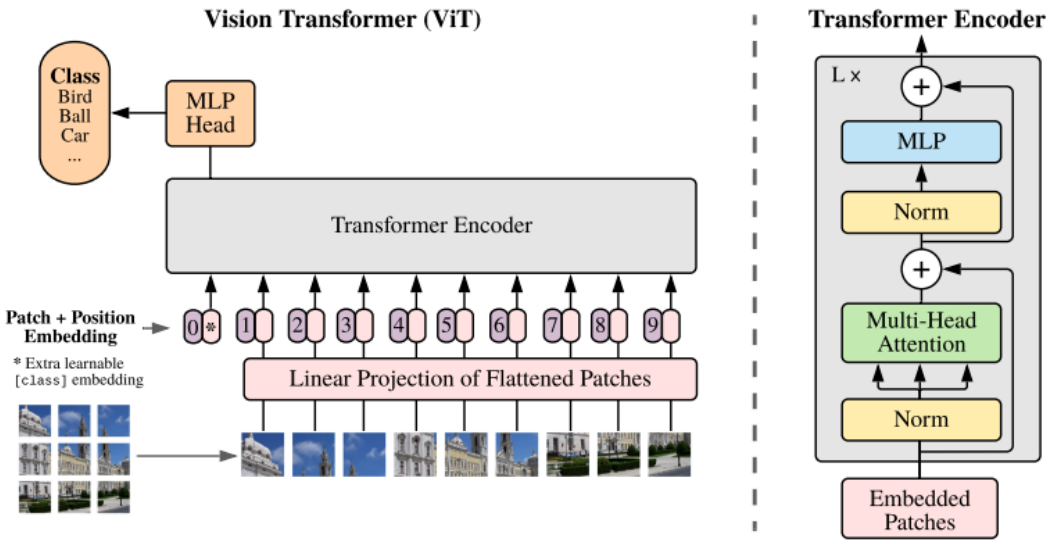
Similarly Multi-Head attentions create attention heads with own weights, apply attention, then concatenate and transform to look at different aspects of relationships in global context.

$$\textbf{MHA}(Q,K,V) = \textbf{Concat}(\textbf{head}_1,\ldots,\textbf{head}_h)W^o$$

- **Feed Forward Neural Network (FFN) –** It's a small Multi Layer Perceptron that processes image patches independently after attention , to learn non-linear patterns.

- **Residual Connections and Layer Norm –** helps to stabilize training and gradients flow, thereby avoiding vanishing gradient problems.

5. **CLS Token and MLP head**
    - At the beginning of patch sequence, a Classification token (**CLS token**) is added that interacts with all other tokens during encoding.
    - It aggregates all information from patches during Transformer encoding, thereby gathering global information.
    - At the end, this token is extracted and passed through another **MLP Head**, (a couple of dense layers and softmax layer for classification). This step gives the predicted label of an image – based on CLS token.



Vision Transformers – Model Architecture

3. METHODOLOGY

3.1 Dataset

- The HAM10000 dataset **[2]**consists of dermatoscopic medical images of seven types of skin lesions as follows:

Actinic keratoses, Basal cell carcinoma, Benign Keratosis, Dermatofibroma, Melanocytic nevi and Vascular Lesions

- The dataset is highly imbalanced with **Melanocytic nevi** significantly having more samples than others.
- This imbalance challenges model training, leading to certain bias towards majority classes.

3.2 Model

- A Vision Transformer (**ViT-Base 32**) model pretrained on ImageNet-21k , a large – scale dataset containing 14 million images across 21,000 classes is chosen from open source – Hugging Face models
- The patch size is 32x32; the sequence of input patches is processed by a stack of 12 Transformer encoder layers.

3.3 Preprocessing and Augmentation

- To prepare the dataset for Vision Transformer, images were resized to 224x224 pixels.
- Pixel values were normalized using mean and standard deviation of ViTImage Processor.
- Data augmentation techniques such as horizontal flipping, small rotations, color jitter were applied to training dataset to improve generalization while preserving the details of medical images.

3.4 Transfer Learning Strategy

- Training a Vision Transformer from scratch requires millions of labeled images and massive computation power.
- However, transfer learning aids to fine tune with fewer labeled images.
- Moreover, it helps to adapt domain – oriented knowledge such as medical imaging, still achieving good performance.
- There are various Transfer Learning strategies widely used in computer vision tasks**.** Let's focus on Two Stage Approach.

Two Stage Approach

- **Stage 1**: Freeze the Vision Transformer and train only the head (head is replaced with a new 7-class output layer).

  This allows the model to adapt pretrained features to the new task

- **Stage 2**: Unfreeze the last few transformer blocks (e.g., Block 10–11–12 in ViT-B/32) and fine-tune them with a much smaller learning rate.

This staged approach helps stabilize training and improves performance even on small datasets.

3.5 Training Setup

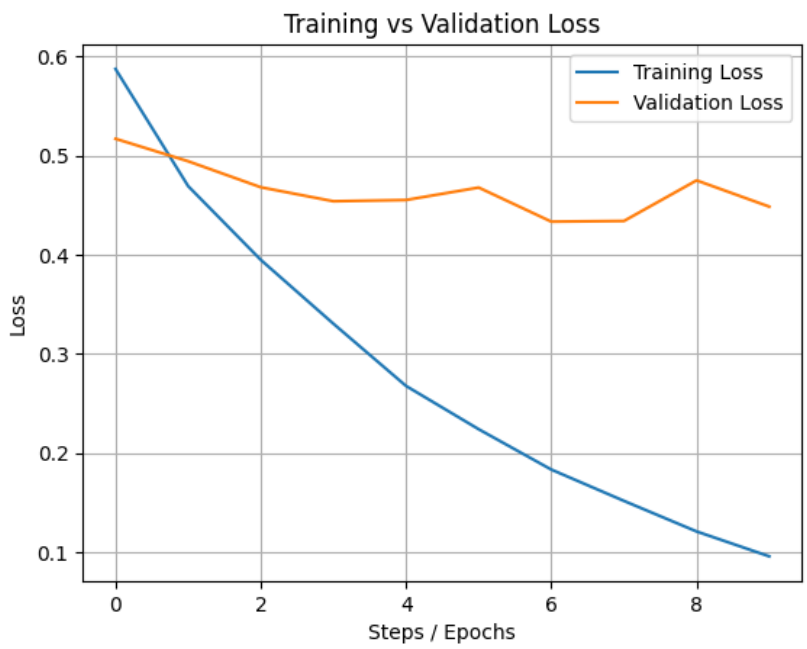
- The dataset contains 10,000 images that were split into training, validation and test datasets (80/10/10).
- In this tutorial, Training was performed using Hugging Face Trainer API.

- The optimizer defaults to **AdamW** and loss function to **cross-entropy loss** for multi-class classification tasks.
- Further, Hyperparameters are listed in the code notebook for reference **[4]**
- Early stopping was applied to prevent overfitting; Evaluation metrics included accuracy and macro F1-score to account for class imbalance.

4. RESULTS

4.1 Training Performance

- Stage 1 – freezing layers except head was trained for 15 epochs that achieved an accuracy of about 0.73 and macro F1 of about 0.35.
- The classifier head alone learns some features but struggles with class imbalance.
- Then Stage 2 – fine tuning was set to 20 epochs , where early stopping happened at 10th epoch as the model was about to overfit , potentially ignored.
- After Fine tuning , **the accuracy is 0.83 with macro-F1 score of about 0.62.**
- Here, fine tuning enables the model to adapt to domain specific patterns while accuracy becomes stable, macro F1 continues to improve, indicating a slight better performance on minority classes compared to stage 1.
- A divergence between training and validation loss was observed , suggesting **over-fitting** – however its being stopped at the 10th epoch.



Fine tuning – Training vs Validation Loss curves

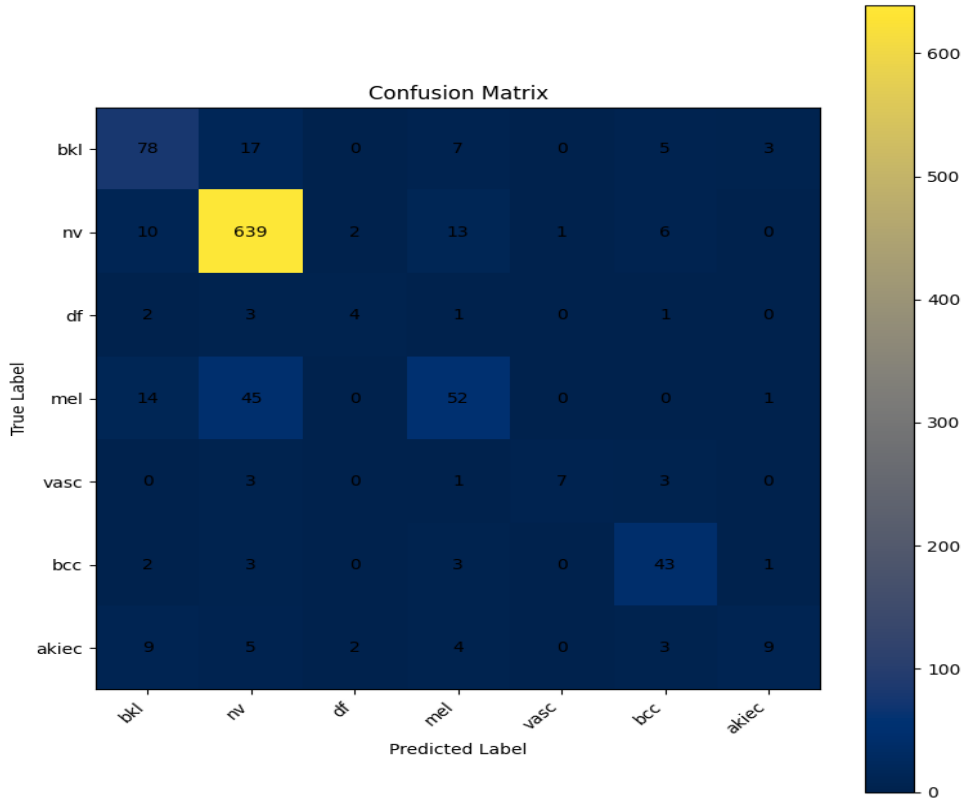
4.2 Classification Results

- The confusion matrix showed that majority classes were classified accurately and minority classes were often mis classified as more frequent classes.

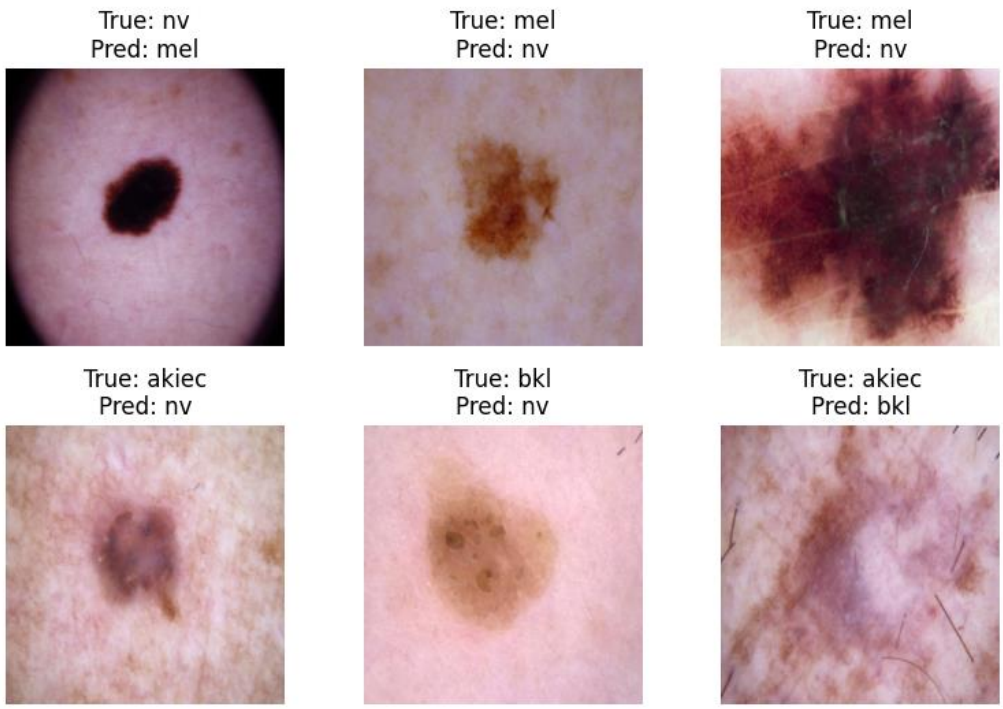
- For example, **Melanocytic nevi (nv)** being the majority class dominates the prediction of certain minority classes such as **Actinic keratoses (akiec).**
- A clear confusion can also be seen between Melanoma and Melanocytic nevi as shown in mis-classified samples.

Improvements

- To address the challenges of dataset class imbalance, future improvements could also be implemented through techniques such as class-weighted loss functions , weighted random sampling for underrepresented classes **.**

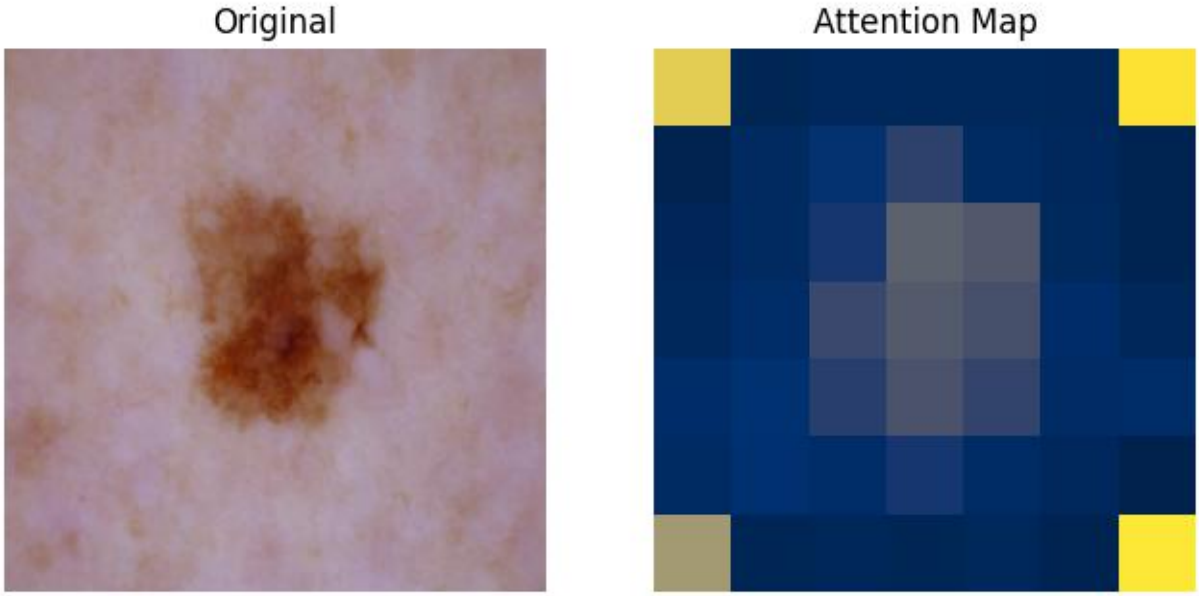


Confusion Matrix – Predicted vs True Labels



Mis-classified samples (True vs Predicted)

4.3 Attention Maps



- To understand the attention focus on images, attention maps were extracted from final transformer layer.
- Attention weights corresponding to CLS token were obtained and averaged across attention heads.
- These values were reshaped into 2D grid structure – to represent spatial distribution of attention across image patches.

- The color scale suggests, **Yellow: high attention (lesion region),** blue: background skin region.
- Additionally, yellow attention is also observed in corner due to global attention where all patches contribute to final predictions (even if importance is relatively low).

5. INTERPRETATION

- The gap between accuracy and macro F1 shows that accuracy alone is not sufficient for evaluating performance in imbalanced datasets. The model achieves good accuracy, however struggles to correctly classify minority classes.
- Data augmentation and Transfer Learning improves generalization and adaptation towards medical imaging, but the model still exhibits some overfitting due to limited data diversity.

6. LIMITATIONS

- Vision Transformers are data hungry in nature. This means that it requires a large dataset for training. However, CNNs can work better in small datasets.
- This is due to concept of **Inductive bias** and **Receptive Field** nature of CNNs.
- This can be understood with an example of solving image puzzles. For small puzzles, almost the nearby pixels – related features are known based on shape/color.
- However for big puzzles , there would be no idea of how image will look like – this requires more training data and time to assemble every part (patches).
- **While this flexibility allows ViTs to model complex global dependencies, it also makes them more data-hungry.**

CONCLUSION

- This tutorial demonstrated the application of Vision Transformers to skin lesion classification using transfer learning.
- The challenges of imbalanced dataset, large data requirements and model performance were discussed. Despite these, the model achieved good results and suggested improvements for better performance.
- The results highlight both potential and challenges of Vision Transformers in medical imaging. They aid in global feature modelling and capture long range spatial relationships if training datasets are huge.
- While this flexibility allows ViTs to model complex global dependencies, it also makes them more data-hungry.

REFERENCES

1. **A. Dosovitskiy *et al.*, "An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2021.**

2. **Ahmad, B. S. Alsulami, and F. Alqurashi, "Enhancing Skin Cancer Detection with Transfer Learning and Vision Transformers," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 10, 2024.**
3. **R. Abla, S. M. Abdelouahed and A. Abdellah, "Fine-Tuning Vision Transformers for Enhanced Skin Lesion Classification: Navigating the Challenges of Small Datasets," *2024 International Conference on Intelligent Systems and Computer Vision (ISCV)*, Fez, Morocco, 2024, pp. 1-5.**
4. **Code File: sri-lekha-19/Machine-Learning-Tutorial: A short demonstration of Vision Transformers on Skin Lesion Classification using Transfer Learning technique**